

Cooperative Aerial Slung Load Transportation Using a Novel Adaptive Sliding Mode Controller

Abhinay N S, Shubhankar Shobhit, Nithya Sridhar, Kaushik Das

Abstract—This paper presents a decentralized approach to transport a cable suspended load using multiple Unmanned Aerial Vehicles (UAVs). The proposed approach is decentralized in the sense that the trajectory for each of the UAVs is computed offline using the kinematics of the system and the desired path of the load. This approach enabled the task of cooperative transportation to be treated as a problem of multiple UAVs, each carrying a cable suspended load. A neural adaptive sliding mode controller proposed by the authors in an earlier work is then deployed on each of the UAVs to track the trajectory while minimizing the load swing. Since the system of an UAV carrying a suspended load is underactuated, the conditions on the sliding surface variables that ensure the stability of the system are also given. The performance of the proposed approach is tested with payloads of non-uniform density and uncertain mass, using Gazebo simulations. The performance is compared against an adaptive sliding mode controller which motivated the design of the proposed controller.

I. INTRODUCTION

The applications for Unmanned Aerial Vehicles (UAVs) have been increasing in the past few decades, of which load transportation is being explored a lot in the recent past [1]. Though both fixed wing and multi-rotor UAVs can be used for this application, Multi-rotor UAVs (MUAVs) are preferred for this task over the fixed wing UAVs due to their Vertical Take-off and Landing (VTOL) capability. The load to be transported can be attached to the UAVs either rigidly [2] or through cables [3]. Rigid attachment of the payload to the UAV body tends to make the system sluggish. On the other hand, though attaching the payload using cables complicates the dynamics of the system by increasing the underactuated degrees of freedom, it does not severely hamper the agile maneuvering capability of the aerial vehicle [4], [5]. Though the MUAVs have the VTOL feature, their payload capacity and the endurance are less when compared to their fixed wing counterparts, thus limiting the distance and the weight of the payload that can be transported. In order to overcome this, multiple MUAVs can be used to transport the load [6], [7]. When using multiple MUAVs, if the payload is attached rigidly to all the MUAVs then the problem of sluggish response of the system may only be aggravated. Hence, attaching the payload using cables can be a more efficient choice for load transportation using multiple MUAVs.

After choosing the type of attachment, it is imperative to make a judicious choice of the control framework and

the UAV controller. When using multiple aerial vehicles to transport a load, two types of control approaches namely, centralized control [8] and decentralized control [9], [10] can be used. In a centralized control approach, the communication between the aerial vehicles and the central node is very important for the system to succeed in its mission whereas in a decentralized control approach, each aerial vehicle operates independently and thus is easily scalable to systems with many aerial vehicles. Also, a decentralized approach is inherently redundant and fault tolerant. Hence, in this work, a decentralized control approach is followed. The decentralization is achieved by providing all the aerial vehicles with a trajectory which is determined using the path to be followed by the payload and the system kinematics. This approach enables the problem of multiple aerial vehicles carrying a single load to be treated as multiple aerial vehicles each carrying a dedicated slung load and travelling together while maintaining the desired formation. Thus, the controller chosen to be employed on all the aerial vehicles should be able to perform trajectory tracking and load swing minimization while being robust to external disturbances and system uncertainties.

Several types of controllers have been proposed to address the problem of slung load transportation using single and multiple MUAVs. A Lyapunov analysis based backstepping controller was proposed in [3] to enable the trajectory tracking for the load suspended from a single aerial vehicle. In [11], the cable was modeled as a serial connection of links and a geometric controller was proposed to stabilize the aerial vehicle. Reinforcement learning was used in [12] to learn swing free trajectories and a model predictive controller was used in [13] to control a multi-rotor with a suspended payload and to navigate through the environment avoiding the obstacles. In [14], a passivity based controller was designed for transporting a suspended load with minimum swing. A control framework for trajectory tracking of a load suspended from two aerial vehicles was presented in [15] wherein the payload was modeled as a point mass. An adaptive controller based on linear quadratic Gaussian method and Lyapunov analysis was proposed in [16] for slung load transportation by multiple aerial vehicles and numerical simulations were given. A feedback linearization based controller was proposed in [17] for carrying a suspended load using three quadrotors. The system of multiple aerial vehicles carrying a load suspended using flexible cables was proven to be a differentially flat system and a controller was synthesized using the differential flatness property in [18]. In [10], model predictive controller was used to transport

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the load using multiple MUAVs in the presence of obstacles and wind disturbance. The work presented in this paper is different from all of the above works with respect to the novel controller used here [19].

The proposed controller was designed by combining adaptive control theory and Super-Twisting Sliding Mode Controller (STSMC) [20], which is a special second order sliding mode controller, accompanied with Radial-Basis Function Neural Networks (RBFNN) [21]. This controller choice was motivated by the fact that Sliding Mode Controller (SMC) is a robust non-linear controller, which is capable of handling external disturbances and system uncertainties, but it suffers from the problem of chattering which is a major hindrance in applying SMC to mechanical systems. The combination of adaptive control theory and higher order sliding mode control is used to address the issue of chattering. RBFNN is used in the proposed control law to provide a continuous approximation of the disturbances which reduces the dependency on SMC to handle them, leading to further reduction in the chattering as will be shown later in the simulation study. RBFNN is chosen over a conventional multi-layer perceptron kind of neural network because the former is a shallow neural network and does not require heavy computation power. Since the system is underactuated, the parameters of the sliding surface play a pivotal role in designing stable sliding surfaces. In this work, trajectory tracking while minimizing the load swing is achieved by deriving the conditions for stable sliding surfaces for the underactuated system motivated by the work presented in [22] and [23]. This work also considers payloads with non-uniform mass distribution because most of the payloads to be transported in the real world fall under this category. The choice of using adaptive control was also motivated by this requirement of carrying non-uniform payloads because for such payloads it is difficult to determine the effective mass of the payload carried by each MUAV.

The contribution of this paper can be summarized as follows. A decentralized approach to solve the problem of slung load transportation using multiple aerial vehicles is proposed. A novel neural adaptive sliding mode controller is used along with stable sliding surfaces specifically designed for transporting a load using a single MUAV, while minimizing the load swing. Simulation studies for a non-uniformly dense payload are presented for the first time to the best of the authors' knowledge.

The rest of the paper is organized as follows. Section II describes the problem formulation followed by the solution methodology explained in Section III. The results obtained from the Gazebo simulations are discussed in Section IV followed by conclusive remarks and future work in Section V.

II. PROBLEM DESCRIPTION

An object that is attached to multiple aerial vehicles using cables as shown in Fig.1 has to be transported along a given path. Following are the assumptions made in this work.

- The load is attached to the MUAVs using inelastic cables and also the attachment points on the MUAVs coincide with their respective Center of Gravity (CoG) points such that the attitude dynamics of the aerial vehicles are decoupled from the attitude dynamics of the load.
- The loads that are to be transported are rigid and the cables are always under tension.
- Complete state of the system is available for feedback.
- All the MUAVs are of the same class.
- Time synchronization is maintained on all the MUAVs.

Considering the advantages of following a decentralized control approach, it was decided to develop a system that can transport a load using multiple MUAVs without the need for explicit communication between the MUAVs or a centralized mission controller. Some of the other desirable characteristics of this multi-MUAV system are highlighted below.

- The system should be capable of handling loads of different size and shape.
- It should be able to track the trajectory in the presence of disturbances and uncertainties.
- In the event of cable breakage and MUAV separation, the remaining agents should complete the mission seamlessly provided the weight of the object does not exceed the combined payload limit of the remaining MUAVs.

III. SOLUTION METHODOLOGY

The representation of a slung load being transported using multiple MUAVs is given in Fig.1, wherein I_x, I_y, I_z represents the inertial frame of reference and B_x^i, B_y^i, B_z^i represents the body frame of reference of the i^{th} MUAV, whose origin is located at the CoG of the corresponding MUAV and B_x^p, B_y^p, B_z^p is the body fixed reference frame of the slung load being transported, whose origin is located at the payload centroid.

In order to make the suspended load follow the given path, the paths for the individual MUAVs have to be determined and the MUAVs are required to follow these paths while minimizing the load swing. The desired paths for the MUAVs can be determined using the kinematics of the system as

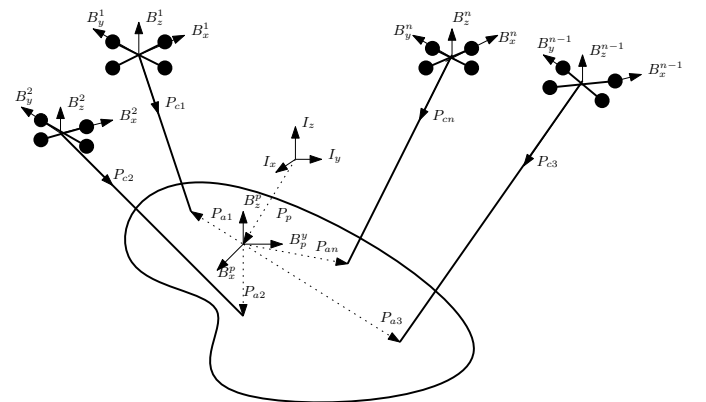


Fig. 1: Load transported by multiple MUAVs

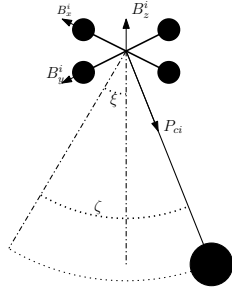


Fig. 2: Cable angles and cable direction

given here,

$$P_i = P_p + R_L^I P_a + R_C^I P_c \quad (1)$$

wherein P_i and P_l represents the desired position of the i^{th} MUAV and the position of the load in the inertial frame respectively. P_a is the position of the attachment point of the cable corresponding to the i^{th} MUAV in the payload frame of reference. R_L^I is the rotation matrix from the payload frame to the inertial frame, which depends on the desired attitude of the payload. P_c is the vector along the cable and R_C^I is the rotation matrix from the cable frame of reference to the inertial reference frame.

As seen from (1), the resulting paths depend on the direction of the cables, which can be determined by the Euler angle pair ζ and ξ (cable angles) as shown in Fig.2. The angles ζ (about y axis) and ξ (about x axis) are measured with respect to a reference frame, H , whose origin coincides with the CoG of the aerial vehicle and the orientation coincides with the inertial frame. The cable direction expressed with respect to the H frame is given by,

$$R_C^I P_c = -l \begin{bmatrix} \sin \zeta & -\sin \xi \cos \zeta & \cos \xi \cos \zeta \end{bmatrix} \quad (2)$$

wherein, l is the length of the cable. So, in order to determine the paths for the aerial vehicles, the desired cable directions have to be decided, which in turn means that the desired values for the cable angles have to be decided. Once the desired values for the cable angles are determined, paths for the individual MUAVs can be computed easily. After the paths are determined, time synchronous trajectories can be designed along those paths in order to make sure that all the MUAVs move in synchronization. This enables the problem of transporting a load using multiple MUAVs to be treated as individual MUAVs each transporting a different load as explained in I. This also provides a decentralized solution to the problem because the aerial vehicles need not communicate with each other or with an external computer since they are all given their individual missions.

Since, the problem is now reduced to trajectory tracking by a single aerial vehicle while minimizing the load swing, an RBFNN based adaptive super-twisting sliding mode controller, proposed by the authors in [19], is used here due to its advantages which were described earlier. The synthesis of the control law for this particular system is presented next. The non-linear dynamics of a single aerial vehicle carrying

a suspended load are as given below,

$$M\ddot{q} + C(q, \dot{q})\dot{q} + G = \tau \quad (3a)$$

$M =$

$$\begin{bmatrix} m_p + m_i & 0 & 0 & 0 & -lm_p c \xi \\ 0 & m_p + m_i & 0 & lm_p c \xi c \zeta & -lm_p s \xi s \zeta \\ 0 & 0 & m_p + m_i & lm_p s \xi c \zeta & lm_p s \xi c \zeta \\ 0 & lm_p c \xi c \zeta & lm_p s \xi c \zeta & l^2 m_p c^2 \zeta & 0 \\ -lm_p c \xi & -lm_p s \xi s \zeta & lm_p s \xi c \zeta & 0 & l^2 m_p \end{bmatrix} \quad (3b)$$

$$C = \begin{bmatrix} lm_p s \xi \dot{\zeta}^2 \\ -lm_p (s \xi c \xi \dot{\zeta}^2 + s \xi c \xi \dot{\zeta}^2 + 2s \xi c \xi \dot{\zeta} \dot{\xi}) \\ lm_p (-2s \xi s \xi \dot{\zeta} \dot{\xi} + c \xi c \xi \dot{\zeta}^2 + c \xi c \xi \dot{\zeta}^2) \\ -l^2 m_p s 2 \xi \dot{\zeta} \dot{\xi} \\ l^2 m_p s \xi c \xi \dot{\zeta}^2 \end{bmatrix} \quad (3c)$$

$$G = g \begin{bmatrix} 0 & 0 & m_p + m_i & lm_p s \xi c \zeta & lm_p c \xi s \zeta \end{bmatrix}^T \quad (3d)$$

$$\tau = \begin{bmatrix} f_x & f_y & f_z & 0 & 0 \end{bmatrix}^T \quad (3e)$$

and the corresponding linearized dynamics are given as,

$$\ddot{x} = \frac{f_x}{m_i} - \frac{m_p g \xi}{m_i} + \chi_x \quad (4a)$$

$$\ddot{y} = \frac{f_y}{m_i} + \frac{m_p g \xi}{m_i} + \chi_y \quad (4b)$$

$$\ddot{\xi} = \frac{-f_y}{m_i l} - \frac{(m_p + m_i) g \xi}{m_i l} + \chi_\xi \quad (4c)$$

$$\ddot{\zeta} = \frac{f_x}{m_i l} - \frac{(m_p + m_i) g \xi}{m_i l} + \chi_\zeta \quad (4d)$$

$$\ddot{z} = \frac{f_z}{m_p + m_i} - g + \chi_z \quad (4e)$$

wherein, $q = [x, \dot{x}, y, \dot{y}, z, \dot{z}, \xi, \dot{\xi}, \zeta, \dot{\zeta}]^T$ represents the state of the system, which includes the position and velocity of the i^{th} aerial vehicle in the inertial frame along with the orientation and the angular rates of the cable attached to the i^{th} aerial vehicle. M is the inertia matrix, C contains the Coriolis terms, G contains the forces due to gravity and f_x, f_y, f_z are the force control inputs which are also defined in the inertial frame. m_i is the mass of the i^{th} aerial vehicle, m_p is the effective mass of the payload that the i^{th} aerial vehicle will be carrying and l is the length of the cable. χ_j , where $j \in \{x, y, \xi, \zeta, z\}$ represents the lumped disturbances due to the neglected higher order dynamics, exogenous disturbances and parametric uncertainties. As can be seen from (4), the translational dynamics along the three axes are independent of each other and thus can be treated as three independent systems. Of these three systems, the equations of motion for the x and y axes suggest that they are underactuated systems with one degree of underactuation. The position of the aerial vehicle is the actuated state variable while the corresponding cable angle is the underactuated state variable. Since the control of these underactuated state variables is necessary to minimize the load swing, the sliding surfaces for these two underactuated systems are designed as a linear combination of the position and the velocity errors of the actuated as well

as the underactuated variables. Since the z-axis system is fully actuated, the corresponding sliding surface is designed as a linear combination of the position and the velocity errors of only the MUAV along the z-axis. The expressions for the sliding surfaces of all the three systems are given as,

$$\sigma_x = \alpha_x \dot{e}_x + \lambda_x e_x + l \alpha_\zeta \dot{e}_\zeta + l \lambda_\zeta e_\zeta \quad (5a)$$

$$\sigma_y = \alpha_y \dot{e}_y + \lambda_y e_y + l \alpha_\xi \dot{e}_\xi + l \lambda_\xi e_\xi \quad (5b)$$

$$\sigma_z = \alpha_z \dot{e}_z + \lambda_z e_z \quad (5c)$$

$$e_j = j - j_{des}, \quad j \in \{x, y, z, \zeta, \xi\} \quad (5d)$$

$$\dot{e}_j = \dot{j} - \dot{j}_{des}, \quad j \in \{x, y, z, \zeta, \xi\} \quad (5e)$$

It is known that, in the case of fully actuated systems, for a linear sliding surface to be stable, the coefficients of that sliding surface have to be positive. It is not necessary that the same condition is sufficient to guarantee the stability of the sliding surfaces for underactuated systems. Following the work presented in [22] and [23], the derivation of the conditions for the stability of the x axis underactuated system is given here. Consider the sliding surface σ_x for the x axis system as given in (5). The derivative of σ_x can be written as,

$$\dot{\sigma}_x = \alpha_x \ddot{x} + l \alpha_\zeta \ddot{\zeta} + \lambda_x \dot{e}_x + l \lambda_\zeta \dot{e}_\zeta \quad (6)$$

Since, this work deals with only the position control of the load and thereby the UAVs, it is assumed that the desired accelerations are zero. Considering the expressions for $\ddot{x}$ and $\ddot{\zeta}$ from the linearized dynamics given in (4), and by defining

$$f_1^x \triangleq -\frac{m_p g \zeta}{m_i} \quad b_1^x \triangleq \frac{1}{m_i} \quad (7a)$$

$$f_2^x \triangleq -\frac{(m_p + m_i) g \zeta}{m_i} \quad b_2^x \triangleq \frac{1}{m_i} \quad (7b)$$

the sliding mode dynamics can be written as,

$$\dot{\sigma}_x = f_1^x \alpha_x + f_2^x \alpha_\zeta + f_x (b_1^x \alpha_x + b_2^x \alpha_\zeta) + \lambda_x \dot{e}_x + l \lambda_\zeta \dot{e}_\zeta \quad (8)$$

The disturbances defined in (4) are not considered in (7) because when in sliding mode, the system is indifferent to the disturbances. The control input f_x is chosen as,

$$f_x = -\frac{f_1^x \alpha_x + f_2^x \alpha_\zeta + \lambda_x \dot{e}_x + l \lambda_\zeta \dot{e}_\zeta}{b_1^x \alpha_x + b_2^x \alpha_\zeta} + \frac{u_{nsmc}^x}{b_1^x \alpha_x + b_2^x \alpha_\zeta}, \quad (9)$$

wherein u_{nsmc}^x is the control input generated by the proposed neural SMC controller in order to drive the system into the sliding mode and it will be defined later in this section. Considering the control signal defined above and considering that $u_{nsmc}^x = 0$ will be zero in an ideal sliding mode, the error dynamics of the cable angle ζ can be written as,

$$\ddot{e}_\zeta = \frac{\alpha_x (f_2^x b_1^x - f_1^x b_2^x)}{l (b_1^x \alpha_x + b_2^x \alpha_\zeta)} - \frac{b_2^x (\lambda_x \dot{e}_x + l \lambda_\zeta \dot{e}_\zeta)}{l (b_1^x \alpha_x + b_2^x \alpha_\zeta)} \quad (10a)$$

$$\ddot{e}_\zeta = \frac{-(\alpha_x g e_\zeta + \lambda_x \dot{e}_x + l \lambda_\zeta \dot{e}_\zeta)}{l (\alpha_x + \alpha_\zeta)} \quad (10b)$$

$$e_\zeta = \zeta - \zeta_{des} = \zeta, \quad \because \zeta_{des} = 0 \quad (10c)$$

Since $\sigma_x = 0$ in the sliding mode, $\dot{e}_x$ can be written as,

$$\dot{e}_x = -\frac{\lambda_x}{\alpha_x} e_x - l \frac{\alpha_\zeta}{\alpha_x} \ddot{e}_\zeta - l \frac{\lambda_\zeta}{\alpha_x} \dot{e}_\zeta \quad (11)$$

Combining (10) and (11), the complete error dynamics of the system in the sliding mode can be written as,

$$\dot{Y} = AY \quad (12a)$$

$$Y = [y_1, y_2, y_3]^T \quad (12b)$$

$$y_1 = e_\zeta, \quad \dot{y}_1 = y_2 = \dot{e}_\zeta, \quad y_3 = e_x \quad (12c)$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ A_{21} & A_{22} & A_{23} \\ -c_3 & -c_2 & -c_1 \end{bmatrix} \quad (12d)$$

$$c_1 = \frac{\lambda_x}{\alpha_x}, \quad c_2 = l \frac{\alpha_\zeta}{\alpha_x}, \quad c_3 = l \frac{\lambda_\zeta}{\alpha_\zeta} \quad (12e)$$

$$A_{21} = -\frac{\alpha_x^2 g + l \lambda_x \lambda_\zeta}{l \alpha_x (\alpha_x + \alpha_\zeta)} \quad (12f)$$

$$A_{22} = \frac{l \lambda_x \alpha_\zeta - l \alpha_x \lambda_\zeta}{l \alpha_x (\alpha_x + \alpha_\zeta)} \quad (12g)$$

$$A_{23} = \frac{\lambda_x^2}{l \alpha_x (\alpha_x + \alpha_\zeta)} \quad (12h)$$

Now, for the error dynamics to be stable, all the eigenvalues of A must have negative real parts which in other words means that A must be Hurwitz. Let $-p_1, -p_2, -p_3$, ($p_1, p_2, p_3 > 0$) be the roots of the characteristic equation of A which defines the desired error dynamics, then the desired characteristic equation can be given as,

$$\lambda^3 + (p_1 + p_2 + p_3) \lambda^2 + (p_1 p_2 + p_2 p_3 + p_3 p_1) \lambda + p_1 p_2 p_3 = 0 \quad (13)$$

Now, determining the characteristic equation of A from its expression given in (12) and comparing it to the characteristic equation given in (13), the sliding surface parameters can be obtained as

$$\lambda_x = \frac{p_1 p_2 p_3}{p_1 p_2 + p_2 p_3 + p_3 p_1} \alpha_x \quad (14a)$$

$$\alpha_\zeta = \frac{\alpha_x (g - (p_1 p_2 + p_2 p_3 + p_3 p_1) l)}{(p_1 p_2 + p_2 p_3 + p_3 p_1) l} \quad (14b)$$

$$\lambda_\zeta = (p_1 + p_2 + p_3) (\alpha_x + \alpha_\zeta) - \lambda_x \quad (14c)$$

As can be seen from (14), if α_x is chosen then all the other parameters of the sliding surface can be computed. Similarly the equations of the sliding surface parameters corresponding to the y axis system can also be computed and the final expressions are given below,

$$\lambda_y = \frac{r_1 r_2 r_3}{r_1 r_2 + r_2 r_3 + r_3 r_1} \alpha_y \quad (15a)$$

$$\alpha_\xi = \frac{\alpha_y (-g + (r_1 r_2 + r_2 r_3 + r_3 r_1) l)}{(r_1 r_2 + r_2 r_3 + r_3 r_1) l} \quad (15b)$$

$$\lambda_\xi = -(r_1 + r_2 + r_3) (\alpha_y - \alpha_\xi) + \lambda_y \quad (15c)$$

wherein, $-r_1, -r_2, -r_3$, ($r_1, r_2, r_3 > 0$) are the desired roots of the characteristic equation for the y axis system. The above

analysis of error dynamics is valid after the system enters the sliding mode. In order to make the system reach and remain on the sliding surface, the neural SMC control input for the x axis system is given as,

$$u_{nsmc}^x = -k_{1x}|\sigma_x|^{0.5}\text{sgn}(\sigma_x) + v_x - \hat{\chi} - \hat{\delta} \quad (16a)$$

$$\dot{v}_x = -k_{2x}\text{sgn}(\sigma_x) \quad (16b)$$

wherein, k_{1x} , k_{2x} are the adaptive gains and $\hat{\chi}$ and $\hat{\delta}$ are the estimation of the approximation of the disturbance by the RBFNN and the estimation of the approximation error respectively. χ is an amalgamation of the lumped disturbances χ_x and χ_z . The expression for the approximation of the disturbance by the RBFNN is given as,

$$\chi = \tilde{W}_x^T \mathbf{h} + \delta \quad (17a)$$

$$\mathbf{h} = \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}\|^2}{\mathbf{b}^2}\right) \quad (17b)$$

wherein, $\tilde{W}_x^T$ is the matrix of optimal weights, $\mathbf{h}$ is the output vector of the hidden layer of the RBFNN, $\mathbf{x}$ is the input vector to the hidden layer, $\mathbf{c}$ is a vector of centers of the nodes of the hidden layer, $\mathbf{b}$ is a vector of variances of the hidden layer nodes and δ is the error in approximation. The expression for the estimation of the approximation of the disturbance is given as,

$$\hat{\chi} = \hat{W}_x^T \mathbf{h} \quad (18)$$

wherein, $\hat{W}_x$ is the matrix of the estimation of the optimal weights. The stability proof for the proposed control law is briefly presented here and the readers are referred to [19] for a detailed proof.

Theorem 1. *The sliding surface σ_x and its derivative $\dot{\sigma}_x$ are finite time stable, provided the control signal is as given in (9), (16) and the adaptation laws for k_{1x} , k_{2x} , $\hat{W}$, $\hat{\mathbf{c}}$, $\hat{\mathbf{b}}$ and $\hat{\delta}$ are as given in (19).*

w_{1x} , w_{2x} , γ_{1x} , γ_{2x} , γ_{3x} , γ_{4x} , γ_{5x} , γ_{6x} are all positive numbers. It should also be noted that k_{1x} , k_{2x} will always remain positive and bounded provided the initial values are chosen to be positive and are updated as given in (19).

$$\dot{k}_{1x} = \begin{cases} w_{1x}\sqrt{\frac{\gamma_{1x}}{2}}, & \text{if } \sigma_x \neq 0 \\ 0, & \text{if } \sigma_x = 0 \end{cases} \quad \dot{k}_{2x} = \begin{cases} w_{2x}\sqrt{\frac{\gamma_{2x}}{2}}, & \text{if } \sigma_x \neq 0 \\ 0, & \text{if } \sigma_x = 0 \end{cases} \quad (19a)$$

$$\dot{\hat{W}}_x = \gamma_{3x}\sigma\mathbf{h}, \dot{\hat{\mathbf{c}}} = \gamma_{4x}\sigma\frac{\partial\mathbf{h}}{\partial\mathbf{c}}\hat{W}_x, \dot{\hat{\mathbf{b}}} = \gamma_{5x}\sigma\frac{\partial\mathbf{h}}{\partial\mathbf{b}}\hat{W}_x, \dot{\hat{\delta}} = \gamma_{6x}\sigma \quad (19b)$$

Proof: Consider a Lyapunov function as given in (20).

$$V = \frac{1}{2}\sigma_x^2 + \frac{1}{2\gamma_{1x}}(k_{1x} - \hat{k}_{1x})^2 + \frac{1}{2\gamma_{2x}}(k_{2x} - \hat{k}_{2x})^2 + \frac{1}{2\gamma_{3x}}\tilde{W}_x^T\tilde{W}_x + \frac{1}{2\gamma_{4x}}\tilde{\mathbf{c}}_x^T\tilde{\mathbf{c}}_x + \frac{1}{2\gamma_{5x}}\tilde{\mathbf{b}}_x^T\tilde{\mathbf{b}}_x + \frac{1}{2\gamma_{6x}}\tilde{\delta}^T\tilde{\delta} \quad (20)$$

wherein, $\tilde{j}$, $j \in \{W_x, \mathbf{c}, \mathbf{b}, \delta\}$ represents the difference between the optimal value and the estimated value.

Let,

$$V_1 = \frac{1}{2}\sigma_x^2 + \frac{1}{2\gamma_{3x}}\tilde{W}_x^T\tilde{W}_x + \frac{1}{2\gamma_{4x}}\tilde{\mathbf{c}}_x^T\tilde{\mathbf{c}}_x + \frac{1}{2\gamma_{5x}}\tilde{\mathbf{b}}_x^T\tilde{\mathbf{b}}_x + \frac{1}{2\gamma_{6x}}\tilde{\delta}^T\tilde{\delta} \quad (21)$$

from which we get,

$$\dot{V}_1 = \sigma_x\dot{\sigma}_x - \frac{1}{\gamma_{3x}}\tilde{W}_x^T\dot{\tilde{W}}_x - \frac{1}{\gamma_{4x}}\tilde{\mathbf{c}}_x^T\dot{\tilde{\mathbf{c}}}_x - \frac{1}{\gamma_{5x}}\tilde{\mathbf{b}}_x^T\dot{\tilde{\mathbf{b}}}_x - \frac{1}{\gamma_{6x}}\tilde{\delta}^T\dot{\tilde{\delta}} \quad (22)$$

Substituting (5a), (4a) and (4d) in (22) the resulting expression becomes

$$\dot{V}_1 = \sigma_x(\alpha_x(f_1^x + b_1^x f_x) + \alpha_z(f_2^x + b_2^x f_x) + \chi) - \frac{1}{\gamma_{3x}}\tilde{W}_x^T\dot{\tilde{W}}_x - \frac{1}{\gamma_{4x}}\tilde{\mathbf{c}}_x^T\dot{\tilde{\mathbf{c}}}_x - \frac{1}{\gamma_{5x}}\tilde{\mathbf{b}}_x^T\dot{\tilde{\mathbf{b}}}_x - \frac{1}{\gamma_{6x}}\tilde{\delta}^T\dot{\tilde{\delta}} \quad (23)$$

Now, choosing f_x as given in (9), (16)

and using (19), (23) can be reduced to

$$\dot{V}_1 \leq -k_{1x}|\sigma_x|^{1.5} \quad (24)$$

Considering (24), the derivative of the Lyapunov function can be written as,

$$\begin{aligned} \dot{V} &\leq -k_{1x}|\sigma_x|^{1.5} + \frac{1}{\gamma_{1x}}(k_{1x} - \hat{k}_{1x})\dot{k}_{1x} \\ &+ \frac{1}{\gamma_{2x}}(k_{2x} - \hat{k}_{2x})\dot{k}_{2x} + \frac{w_{1x}}{\sqrt{2}\gamma_{1x}}|k_{1x} - \hat{k}_{1x}| - \frac{w_{1x}}{\sqrt{2}\gamma_{1x}}|k_{1x} - \hat{k}_{1x}| \\ &\quad - \frac{w_{2x}}{\sqrt{2}\gamma_{2x}}|k_{2x} - \hat{k}_{2x}| - \frac{w_{2x}}{\sqrt{2}\gamma_{2x}}|k_{2x} - \hat{k}_{2x}| \end{aligned} \quad (25)$$

wherein, $\hat{k}_{1x}$ and $\hat{k}_{2x}$ are the upper bounds of the adaptive gains. Since, w_{1x} , w_{2x} , γ_{1x} , γ_{2x} are all positive and $k_{1x} < \hat{k}_{1x}$, $k_{2x} < \hat{k}_{2x}$ as shown in [24], the derivative of the Lyapunov function can be further written as,

$$\begin{aligned} \dot{V} &\leq -k_{1x}|\sigma_x|^{1.5} - |k_{1x} - \hat{k}_{1x}|\left(\frac{\dot{k}_{1x}}{\gamma_{1x}} - \frac{w_{1x}}{\sqrt{2}\gamma_{1x}}\right) \\ &\quad - |k_{2x} - \hat{k}_{2x}|\left(\frac{\dot{k}_{2x}}{\gamma_{2x}} - \frac{w_{2x}}{\sqrt{2}\gamma_{2x}}\right) \end{aligned} \quad (26)$$

If, $\dot{k}_{1x}$, $\dot{k}_{2x}$ are chosen as given in (19), then (26) can be written as,

$$\dot{V} \leq -k_{1x}|\sigma_x|^{1.5} < 0 \quad (27)$$

thus proving the stability of the controller.

Similarly, the expressions for the force control inputs of the y-axis and z-axis systems can be derived.

The attitude and the thrust control inputs for the aerial vehicles can be determined from these force control inputs as

$$T = \sqrt{f_x^2 + f_y^2 + f_z^2} \quad (28a)$$

$$\theta_{des} = \tan^{-1}\left(\frac{f_x \cos(\psi_{des}) + f_y \sin(\psi_{des})}{f_z}\right) \quad (28b)$$

$$\phi_{des} = \tan^{-1}\left(\frac{(f_x \sin(\psi_{des}) - f_y \cos(\psi_{des})) \cos(\theta_{des})}{f_z}\right) \quad (28c)$$

IV. RESULTS

The results obtained from the Gazebo simulations of the proposed approach are presented in this section. Hector quadrotor was used as the MUAV for the simulations. The mass of this MUAV is 2 kg and it can carry a payload of 1 kg. In order to simulate the inelastic cables, massless cylindrical rods were used. These rods were attached to the aerial vehicles using two revolute joints, with their axis of rotation orthogonal to each other, in order to mimic the type of joint when actual cables will be used in the real world. In order to demonstrate the performance of the proposed approach while carrying different shapes of loads using cables of different lengths, simulations were carried out with four MUAVs carrying 1D (beam) and 2D (sheet) loads. The simulations were carried out with loads of non-uniform mass density. In order to simulate loads with non-uniform mass density, the load was divided into four geometrically identical parts. Each of these parts were given a different mass and finally all these parts were attached together using fixed joints in Gazebo. For all the cases, it was assumed that the given path to be followed by the payload was a square of side 20 m at a constant altitude of 5 m and the paths for the aerial vehicles were determined using the kinematics as mentioned in III. In all the simulations, the desired orientation of the payload was assumed to be coinciding with the orientation of the inertial frame so that the rotation matrix R_L^I is an identity matrix and also the desired cable angles are taken to be 0 so that the cables remain vertical. Once the paths for the MUAVs were computed, linear trajectories with a speed of 1 m/s were designed along those paths and were given as inputs to the MUAVs. The UAVs were given a time of 30 s to take off and correct the orientation of the cables before starting the trajectory tracking. During all these simulations, the systems were tested with a payload mass uncertainty of 50%. Also to test the ability of the controller in handling non-uniformly dense payloads, the effective mass of the payload carried by each of the MUAV, which is an input to the control law, was computed assuming a uniform mass distribution and dividing the mass of the payload equally among all the MUAVs.

The performance of the proposed control law for a system of four MUAVs carrying a 2D load is compared with the performance of the adaptive control law [24], which will be referred to as the original adaptive control law hereafter, based on which the proposed law is designed. The original adaptive control law for the x axis system is given as

$$\ddot{u}_{smc}^x = -k_{1x}|\sigma_x|^{0.5}\text{sgn}(\sigma_x) + \dot{v}_x \quad (29a)$$

$$\dot{v}_x = -k_{2x}\text{sgn}(\sigma_x) \quad (29b)$$

and the corresponding update laws are given as

$$\dot{k}_{1x} = \begin{cases} w_1 \sqrt{\frac{\eta_1}{2}}, & \text{if } \sigma_x \neq 0 \\ 0, & \text{if } \sigma_x = 0 \end{cases}, \quad (30a)$$

$$k_{2x} = 2\epsilon k_{1x} + \lambda + 4\epsilon^2, \quad (30b)$$

wherein ϵ and λ are arbitrary positive constants. Similarly one can obtain the expressions for the y and z axes systems.

These comparison results are followed by the results obtained from a four MUAV system while carrying a 1D load. The sliding surface parameters used in the simulations are given below,

$$\alpha_x = 1.0, \lambda_x = 0.333, \alpha_z = 0.308, \lambda_z = 3.591 \quad (31a)$$

$$\alpha_y = 1.0, \lambda_y = 0.333, \alpha_\xi = -0.308, \lambda_\xi = -3.591 \quad (31b)$$

$$\alpha_y = 2.0, \lambda_y = 2.0 \quad (31c)$$

These parameters were arrived at by trial and error approach. The dynamics of the single aerial vehicle with a cable suspended load were numerically simulated and different desired eigenvalues of the A matrix were tested. The chosen values maintained a good balance between reducing the position error as well as the cable orientation error. The inputs to the three RBFNNs were the corresponding sliding variables. The parameters of the RBFNNs were chosen to be same for the x, y and z axes systems. The weights of the RBFNN were initialized with zeros. The range of the centers and the variances was decided based on the expected range of the sliding surface. 40 nodes were considered in the hidden layer with centers belonging to $[-0.5, 0.5]$ spaced at an equal interval of 0.025. The variances of all the hidden nodes was taken as 2.0. The values $\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5, \gamma_6$ were all taken as 0.1. The initial values of the adaptive gains of the proposed control law were also same for all the three (x, y, z) systems and are given below,

$$k_1(0) = 0.1 \quad (32a)$$

$$k_2(0) = 0.1 \quad (32b)$$

. The parameters corresponding to the original control law for all the three (x, y, z) systems were chosen as

$$\epsilon = 1, \lambda = 1, k_1(0) = 0.1$$

$$k_2(0) = 2\epsilon k_1(0) + \lambda + 4\epsilon^2, \gamma_1 = 1.0$$

A. Four MUAV System carrying a 2D load

A square sheet of side 2 m was used as the load for the simulations as shown in Fig.3. The mass of the load was 2.0 kg and the length of the cables used was 2.5 m. To simulate the non-uniform payload, the loads were divided into four squares of side 1 m with masses as 0.25 kg, 0.25 kg, 0.75 kg and 0.75 kg. The initial positions of the four MUAVs were (1, 8.5), (-1, 8.5), (1, 1.5) and (-1, 1.5). Since the rods used to represent the cables cannot be collapsed they had to be attached to the MUAVs at an angle to properly start the simulation, hence the initial large values of the cables' orientation about one of the axes. The errors in positions of the four MUAVs, orientation of the cables and the control forces while implementing the proposed control law are given in Fig.4, Fig.5 and Fig.6 respectively. The errors in position of the MUAVs, the cable orientations and the control forces corresponding to the original adaptive law are given in Fig.7, Fig.8 and Fig.9 respectively. During the takeoff phase, it can be seen from the above figures that the original adaptive law is able to keep the ξ angle of the cables within tighter bounds when compared to the proposed law but the

performance of both the laws is similar with respect to the ζ angle control. During the trajectory tracking phase, the performance of both the laws with respect to the ξ angle is also very similar. From the position error plots it can be seen that the maximum error in the x and y axes when using the proposed law is about 0.5 m greater when compared to the original adaptive law and again the performance of both the laws with respect to the z axis is very much similar. It is clearly evident from the control signal plots that the original adaptive law achieves marginally better performance at the cost of high chattering in the control signals which is not desirable, whereas the chattering present in the proposed control law is relatively very less. So the proposed control law is more suitable to be implemented on mechanical systems like the MUAVs which have actuators with a limited bandwidth.

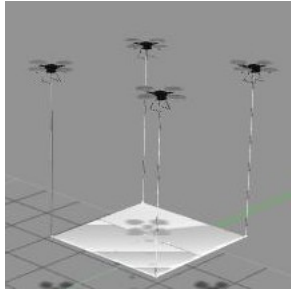


Fig. 3: Four MUAVs carrying a 2d load

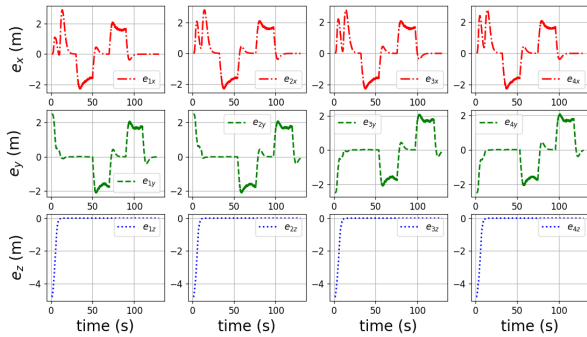


Fig. 4: Error in positions of all the four MUAVs along the x, y and z axes carrying a 2D load while implementing the proposed law

B. Four MUAV system carrying a 1D load

A long rod of length 4 m was attached to the four MUAVs using cables of length 5.0 m as shown in Fig.10. The mass of the load was 2.0 kg and it was divided into four identical parts as done in the above case. Here the results obtained from the proposed law are only presented. The positional error plots are given in Fig.11. A large error in the positions of the MUAVs can be seen in the x and y axes during the takeoff phase but the system recovers before the completion of the time allotted for the takeoff phase. This large initial

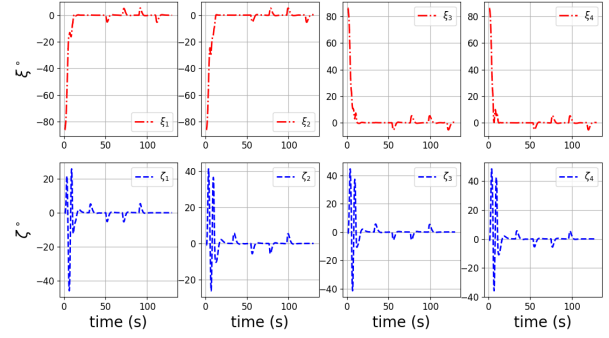


Fig. 5: Orientation of the cables of all the four MUAVs represented by ζ and ξ carrying a 2D load while implementing the proposed law

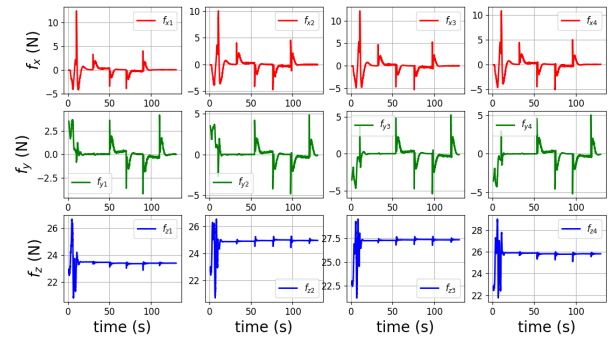


Fig. 6: Force control signals of all the four MUAVs along the x, y and z axes carrying a 2D load while implementing the proposed law

error can be attributed to the non-optimal selection of the sliding surface parameters. It can be seen that the maximum error in position for all the MUAVs during the trajectory tracking phase does not exceed 2 m. From the Fig.12, it can be seen that the load swing is minimized by maintaining the cable angles within a tight bound close to zero and from the Fig.13 it can be seen that the chattering is very less similar to the previous case.

From the positional error plots presented for the two cases, it can be seen that there is periodic increase and decrease in the error along the x and y subsystems. This can be attributed to the sudden change in the desired velocity while transitioning from one side of the square to the other (Once every 20 s, since the side of the square was chosen as 20 m and the speed along the trajectory was chosen as 1 m/s). Since it has been shown that the proposed controller was able to minimize the load swing while tracking the trajectory with reasonable positional errors, it can be concluded that the payload was also following the desired path with acceptable accuracy. While deriving the expressions for the sliding surface parameters, it was stated that u_{nsmc}^x will be zero in an ideal sliding mode. To check the correctness of the above statement, the plots of the sliding variables and the plots of

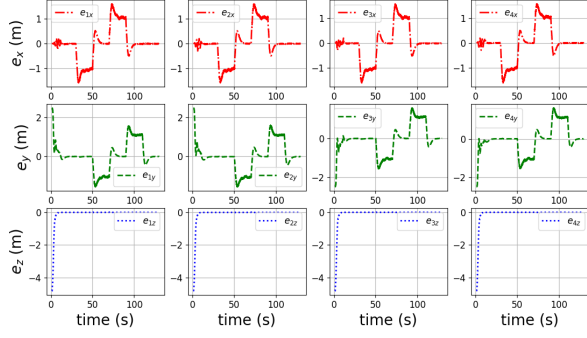


Fig. 7: Error in positions of all the four MUAVs along the x,y and z axes carrying a 2D load while implementing the original adaptive law

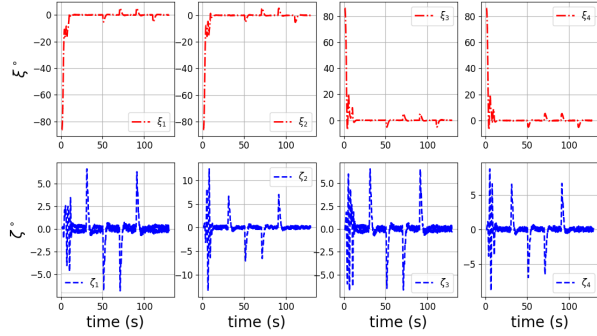


Fig. 8: Orientation of the cables of all the four MUAVs represented by ζ and ξ carrying a 2D load while implementing the original adaptive law

the contribution by the RBFNNs for the x and y subsystems for the four MUAV system carrying a load with cables of length 2.5 m are given in Fig.14 and Fig.15 respectively.

It can be seen from those two figures that the sliding variables value increase only when there is a transition from one side of the square to the other side which is in line with the characteristics observed in the positional error plots and the contributions by the RBFNN are very small throughout the simulation. From these two observations, it can be concluded that when in sliding mode u_{nsmc}^x will be close to 0 since it is a sum of the RBFNN contribution and a term that depends on the magnitude of the sliding variable. It won't be exactly 0 because ideal sliding modes do not exist in practical systems. A similar observation can be made about u_{nsmc}^y which is the neural SMC part of the control signal for the y axis system. From all the simulation cases presented above, the capability of the controller in tracking the trajectory while minimizing the load swing and the chattering in the control signals can be established. The simulation videos of all the different cases presented in this paper and also some additional case are combined into a single video and is available at [25].

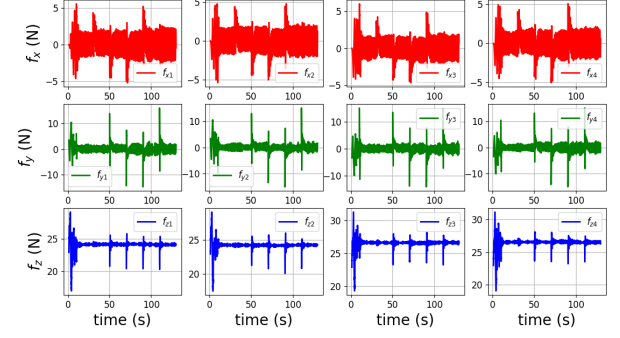


Fig. 9: Force control signals of all the four MUAVs along the x,y and z axes carrying a 2D load while implementing the original adaptive law

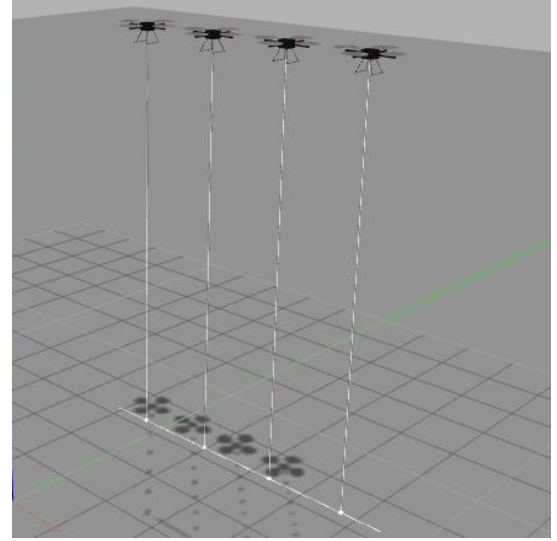


Fig. 10: Four MUAVs carrying a 1D beam like load

V. CONCLUSION AND FUTURE WORK

In this paper, a decentralized approach for transporting a slung load using multiple aerial vehicles is proposed and validated using extensive simulations. The capability of the designed controller in handling non-uniformly dense payload is established. It can also be concluded that there is no need to dynamically determine the effective mass of the payload carried by each MUAV as the adaptive nature of the proposed controller is shown to be capable of handling payload mass uncertainties. Since the proposed approach is decentralized, it can be scaled up, to carry heavier loads with the required number of aerial vehicles. This work only shows that the proposed approach is capable of handling the cooperative aerial slung load transportation task but the obtained performance is in no terms an optimal performance. This is due to the selection of the sliding surface and controller parameters using a trial and error method. A series of research activities are planned as part of the future work that needs to be done in order to deploy the proposed approach in the real world. Some of those activities include determining

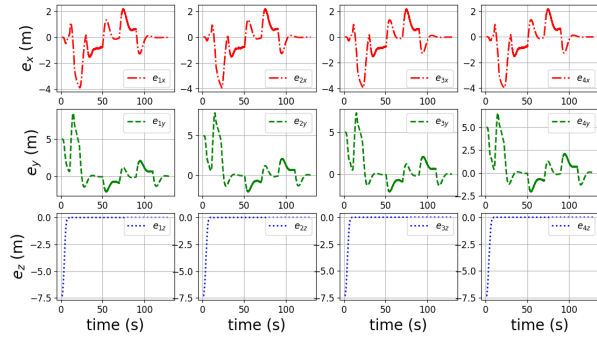


Fig. 11: Error in positions of all the four MUAVs along the x,y and z axes carrying a 1D load while implementing the proposed law

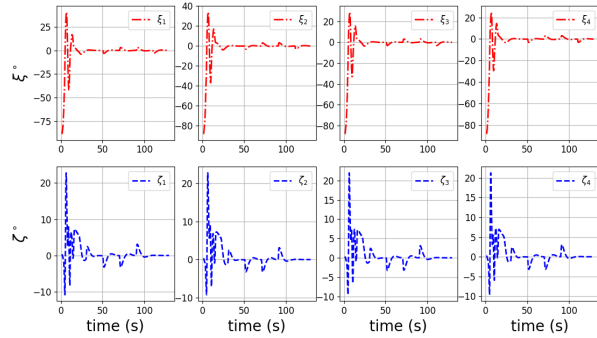


Fig. 12: Orientation of the cables of all the four MUAVs represented by ζ and ξ carrying a 1D load while implementing the proposed law

an analytical solution to select the optimal sliding surface parameters along with an in-depth analysis of the effect of the various control parameters to obtain an optimal performance, studying the effect of measurement noise and wind gust on the performance of the controller, implementing collision avoidance and carrying the load using heterogeneous MUAVs.

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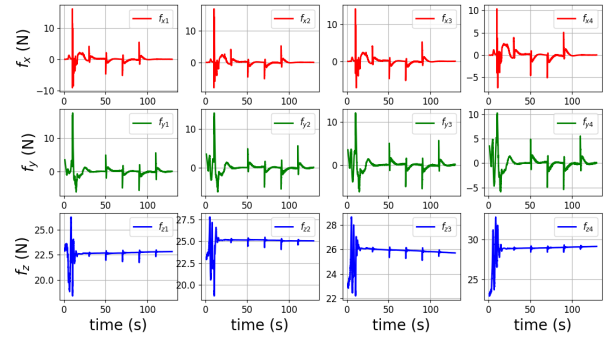


Fig. 13: Force control signals of all the four MUAVs along the x,y and z axes carrying a 1D load while implementing the proposed law

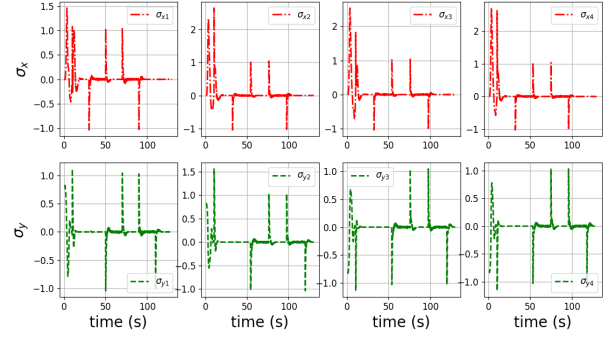


Fig. 14: Time evolution of the sliding variables of the x and y subsystems carrying a 2D load suspended from four MUAVS with cables of length 2.5 m while implementing the proposed law

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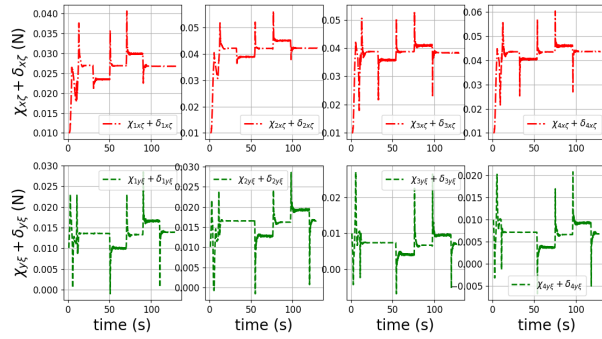


Fig. 15: Contribution by RBFNN for the x and y subsystems carrying a 2D load suspended from four MUAUVs with cables of length 2.5 m while implementing the proposed law

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